Customer Shopping Behavior Analysis

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1. Project overview

A retail company wants to know how its customers shop: who spends the most, which products and categories bring in revenue, and whether discounts and the subscription program are actually paying off. This project answers those questions from 3,900 purchase records, which together come to $233,081 in revenue.

The work ran in three stages. Python (pandas) was used to clean the data and add a few derived columns. The cleaned table was then loaded into PostgreSQL and queried with SQL to answer ten business questions. Finally, Power BI was used to build an interactive dashboard that management can filter by subscription status, gender, category and shipping type.

The main objectives were:

- Find out which customer groups (gender, age group, subscribers) bring in the most revenue
- Rank products and categories by sales, ratings and discount use
- Check whether subscriptions, discounts and faster shipping lead to bigger purchases
- Turn the findings into recommendations the business can act on

2. Dataset summary

- Rows: 3,900 (one row per customer purchase)
- Columns: 18
- Key features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method)
- Missing data: 37 values in the Review Rating column

A few numbers give a sense of scale. Customers are between 18 and 70 years old, with a median age of 44. Purchase amounts range from $20 to $100 and average $59.76. The data covers 25 items across 4 categories, sold in all 50 US states, and the average review rating is 3.75 out of 5.

3. Exploratory data analysis using Python

We began with data preparation and cleaning in Python:

- Data loading: imported the dataset using pandas.
- Initial exploration: used `df.info()` to check structure and `.describe()` for summary statistics.

```
In [3]: df.describe(include="all").T
```

column	count	unique	top	freq	mean	std	min	50%	max
Customer ID	3,900.00				1,950.50	1,125.98	1.00	1,950.50	3,900.00
Age	3,900.00				44.07	15.21	18.00	44.00	70.00
Gender	3900	2	Male	2652					
Item Purchased	3900	25	Blouse	171					
Category	3900	4	Clothing	1737					
Purchase Amount (USD)	3,900.00				59.76	23.69	20.00	60.00	100.00
Location	3900	50	Montana	96					
Size	3900	4	M	1755					
Color	3900	25	Olive	177					
Season	3900	4	Spring	999					
Review Rating	3,863.00				3.75	0.72	2.50	3.80	5.00
Subscription Status	3900	2	No	2847					
Shipping Type	3900	6	Free Shipping	675					
Discount Applied	3900	2	No	2223					
Promo Code Used	3900	2	No	2223					
Previous Purchases	3,900.00				25.35	14.45	1.00	25.00	50.00
Payment Method	3900	6	PayPal	677					
Frequency of Purchases	3900	7	Every 3 Months	584					

The summary already shows two things worth knowing before any analysis. Male customers make up 2,652 of the 3,900 rows (68%), and only 3,863 rows have a review rating, so 37 are missing.

- Missing data handling: the 37 missing Review Rating values were filled with the median rating of the product's own category, so a missing footwear rating gets the footwear median and not the overall one.

```
df["Review Rating"] = df.groupby("Category")["Review
Rating"].transform(lambda x: x.fillna(x.median()))
```

`df.isnull().sum()` before vs after imputation

column	missing before	missing after
Customer ID	0	0
Age	0	0
Gender	0	0
Item Purchased	0	0
Category	0	0
Purchase Amount (USD)	0	0
Location	0	0
Size	0	0
Color	0	0
Season	0	0
Review Rating	37	0
Subscription Status	0	0
Shipping Type	0	0
Discount Applied	0	0
Promo Code Used	0	0
Previous Purchases	0	0
Payment Method	0	0
Frequency of Purchases	0	0

- Column standardization: renamed columns to lowercase snake_case (for example `Purchase Amount (USD)` became `purchase_amount`) so they are easier to query in SQL.
- Feature engineering:
  - Created an `age_group` column by splitting age into four equal-sized bins with `pd.qcut`.

- Created a `purchase_frequency_days` column by mapping text like "Weekly" or "Quarterly" to a number of days.

`age_group created with pd.qcut(df["age"], q=4)`

age_group	min age	max age	customers
Young_Adult	18	31	1028
Adult	32	44	942
Middle_Aged	45	57	986
Senior	58	70	944

- Data consistency check: `discount_applied` and `promo_code_used` turned out to be identical on every row, so `promo_code_used` was dropped as redundant.
- Database integration: the cleaned DataFrame was loaded into a PostgreSQL table called `customer` using SQLAlchemy, ready for the SQL analysis.

```
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:
{port}/{database}")
df.to_sql("customer", engine, if_exists="replace", index=False)
```

One issue came up while re-checking the notebook. The frequency mapping uses the keys "Every 3 months" and "Bi-weekly", but the data spells them "Every 3 Months" and "Bi-Weekly". Because of the capital letters, 1,131 rows (29%) ended up with an empty `purchase_frequency_days`. None of the ten SQL questions use that column, so the results below aren't affected, but the mapping should be fixed before the column is used anywhere.

4. Data analysis using SQL (business transactions)

We ran ten queries against the `customer` table in PostgreSQL. Each one answers a specific business question. The results below were re-checked against the cleaned data.

Q1. Revenue by gender

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Gender	Revenue (USD)	Purchases	Avg purchase (USD)
Male	157,890	2,652	59.54
Female	75,191	1,248	60.25

Men bring in 68% of revenue, but that's because there are more male customers. Women actually spend slightly more per purchase.

Q2. High-spending discount users

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

839 customers used a discount and still spent at or above the $59.76 average. That's half of the 1,677 discount users, so discounts aren't only attracting bargain hunters.

Q3. Top 5 products by average review rating

```
SELECT item_purchased, AVG(review_rating) AS avg_review_rating
FROM customer
GROUP BY item_purchased
ORDER BY avg_review_rating DESC
LIMIT 5;
```

Rank	Product	Avg rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80

Rank	Product	Avg rating
5	Skirt	3.78

The spread is narrow. Even the lowest-rated product (Shirt, 3.62) is only 0.24 points behind Gloves, so no single product has a satisfaction problem.

Q4. Standard vs Express shipping

```
SELECT shipping_type, AVG(purchase_amount) AS avg_purchase_amount
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Express orders average $60.48 against $58.46 for Standard, a gap of about $2 (3.4%). Customers who pay for faster shipping spend a little more, but the difference is small.

Q5. Do subscribers spend more?

```
SELECT subscription_status,
       AVG(purchase_amount) AS avg_spend,
       SUM(purchase_amount) AS total_revenue,
       COUNT(*) AS total_purchases
FROM customer
GROUP BY subscription_status
ORDER BY avg_spend DESC;
```

Subscribed	Customers	Total revenue (USD)	Avg spend (USD)
No	2,847	170,436	59.87
Yes	1,053	62,645	59.49

No, they don't. Subscribers spend about the same per purchase as everyone else, and they make up only 27% of customers. Right now the subscription isn't lifting basket size at all.

Q6. Products with the highest discount rate

```
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0
END) / COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Product	Discount rate
Hat	50.0%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%

About half of all Hat and Sneakers sales use a discount, compared with 43% across the whole store. These items may be selling mainly because of the discount.

Q7. Customer segments (New, Returning, Loyal)

```
SELECT CASE
         WHEN previous_purchases = 0 THEN 'New'
         WHEN previous_purchases BETWEEN 1 AND 10 THEN 'Returning'
         ELSE 'Loyal'
       END AS customer_segment,
       COUNT(DISTINCT customer_id) AS customer_count
FROM customer
GROUP BY 1
ORDER BY customer_count DESC;
```

3,116 customers (80%) are Loyal with more than 10 previous purchases, and 784 are Returning. The New segment is empty because every customer in the data has at least one previous purchase, so this dataset can't tell us anything about first-time buyers.

Q8. Top 3 products in each category

```
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
         ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
DESC) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

Category	#1	#2	#3
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	none

Outerwear only has two products, which partly explains why it's the smallest category. Some ranks are ties (Blouse and Pants both have 171 orders), and `ROW_NUMBER` breaks them arbitrarily.

Q9. Are repeat buyers more likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Of the 3,476 repeat buyers (more than 5 previous purchases), 958 subscribe and 2,518 don't. That's 27.6%, almost exactly the 27% rate for all customers. Buying often doesn't make someone more likely to subscribe, which means the program isn't converting its most obvious audience.

Q10. Revenue by age group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

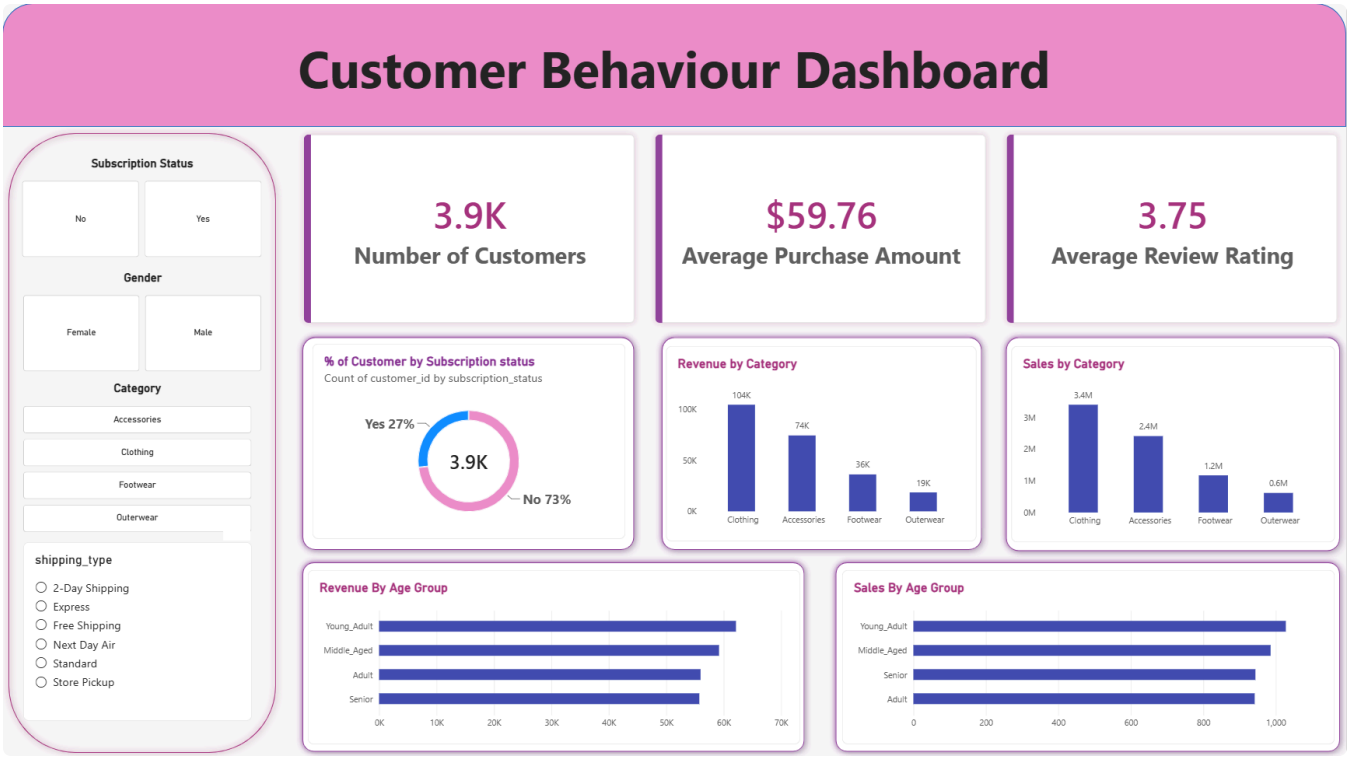
Age group	Ages	Revenue (USD)
Young_Adult	18 to 31	62,143
Middle_Aged	45 to 57	59,197
Adult	32 to 44	55,978
Senior	58 to 70	55,763

Young adults bring in the most revenue, about $6,400 more than seniors. They're also the largest group (1,028 customers), so the gap comes from headcount more than from higher spending.

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI so the findings can be explored without writing queries. It sits on a single page and has three parts:

- Slicers on the left for Subscription Status, Gender, Category and Shipping Type. Every chart on the page updates when one is selected.
- KPI cards at the top: 3.9K customers, $59.76 average purchase and a 3.75 average review rating.
- Five charts: share of customers by subscription status, revenue and sales by category, and revenue and sales by age group.



The category charts make the biggest point on the page. Clothing alone brings in $104,264 (45% of revenue) from 1,737 sales, while Outerwear brings in $18,524 (8%) from 324 sales. Putting revenue and sales count side by side also shows that the average order value is about the same in every category ($57 to $60), so the categories differ in how many people buy, not in how much each order is worth.

6. Business recommendations

1. Rework the subscription program. Subscribers spend $59.49 per purchase against $59.87 for non-subscribers, and repeat buyers subscribe at the same 27% rate as everyone else. The current offer gives people no reason to join or to spend more once they have. Test perks that are tied to spending, such as free express shipping above a minimum order, and target the 2,518 repeat buyers who haven't subscribed yet.
2. Check whether discounts are needed on the most-discounted items. Hat, Sneakers and Coat are sold on discount about half the time. Since 839 discount users already spend above average, try reducing discounts on these items for a few weeks and watch whether their sales actually drop.
3. Grow Outerwear. It has only two products (Jacket and Coat) and earns 8% of revenue. Adding a few products and promoting them in Fall and Winter is a cheap way to grow a category with plenty of room.

4. Keep Clothing and Accessories well stocked. Together they bring in 77% of revenue. Blouse, Pants, Shirt and Jewelry are the top sellers and should be the last items allowed to run out.
5. Market to young adults (18 to 31) and middle-aged customers (45 to 57). These two groups bring in the most revenue. Use the age-group slicer in the dashboard to check which categories each group prefers before planning campaigns.
6. Promote express shipping at checkout. Express orders are about $2 larger on average. Offering it free above a spending threshold could raise order values slightly.

7. Limitations and tools used

A few things in the data limit how far these findings can be pushed:

- Every subscriber and every discount user in the dataset is male. Not one of the 1,248 female customers has a subscription or used a discount. That's almost certainly how the dataset was generated rather than how real customers behave, so the gender results in Q1, Q2 and Q5 should be read with care.
- The data has no purchase dates, so trends over time can't be measured. Season is the only time-related column.
- Every customer has at least one previous purchase, so the "New" segment in Q7 is always empty.
- The `purchase_frequency_days` column is 29% empty because of the capitalization bug described in section 3.

Stage	Tool	Used for
Cleaning and feature engineering	Python (pandas) in Jupyter Notebook	Loading, exploring, imputing missing values, creating new columns
Storage and querying	PostgreSQL, SQLAlchemy, psycopg2	Holding the cleaned table and running the 10 business queries
Visualization	Power BI Desktop	Interactive dashboard with slicers and KPI cards

The project took a raw CSV of 3,900 purchases all the way to a working dashboard. The clearest result is that the subscription program and discounts aren't changing how much people spend, which makes them the first things to fix. A dataset with purchase dates

and a real mix of new and returning customers would let the next round of analysis look at trends and retention as well.